

retains reward model

drops critic

RLHF

2017–2022 · preference-based

Baseline recipe. SFT → reward model learned from human preferences → RL against that learned model.

SFT initialisation

+ learned reward model r_ψ

+ value critic V_ϕ (baseline)

+ on-policy rollout loop

KL penalty to frozen π_{ref}

PPO

2017 optimiser · at scale 2022

Δ vs. RLHF: keeps the reward model and the critic; **adds** a clipped surrogate and GAE advantages.

+ clipped ratio, $\varepsilon \approx 0.2$

+ GAE advantages ($\lambda=0.95$)

4 models: policy, ref, reward, value

GRPO

2024–2025 · group-relative

Δ vs. PPO: **removes** the learned critic; **adds** a group-relative baseline and a verifier reward.

– learned value network (critic)

+ group of G completions / prompt

+ within-group standardisation A_i

+ programmatic verifier (RLVR)

4 → 3 models (no value network)

$\sigma_g = 0 \Rightarrow A_i = 0$: the ZVF point

orthogonal branch: same preference data, no RL loop

DPO

2023 · orthogonal branch

Δ vs. RLHF: reparameterises the KL-constrained objective in terms of the policy; **removes** the reward model and the RL loop.

+ reward via closed-form π_θ

– learned reward model

– value critic

– on-policy sampling / RL loop

classification loss on pref. pairs

no online exploration, no verifier

Chip colour

added

removed

retained

consequence

solid frame: policy-gradient lineage

dashed frame: orthogonal branch